THE FINITE TYPE MODULES ON A RINGED SPACE ARE REPRESENTED UP TO ISOMORPHISM BY A SET

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of a sheaf on a space by glueing data relative to an open covering. Invoked inside the counted proof (source lines 1188-1200) as a cross-chapter pointer. Turns the collection of modules generated on the members of a fixed covering into a collection parametrised by subsheaves and glueing data.
- The definition of a module of finite type as one locally a quotient of a finite direct sum of the structure sheaf. The author's section on modules of finite type, in which the counted statement occurs. Identifies the modules to be represented with those generated on the members of some covering.

Source reference. Original reference sheaves-section-glueing-sheaves: sources/stacks-modules/pinned-original-source.tex, source lines 1199–1199. Navigation pointer only: the counted text refers to the Stacks item sheaves-section-glueing-sheaves, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a ringed space. There exists a set of $\mathcal{O}_X$-modules $\{\mathcal{F}_i\}_{i\in I}$ of finite type such that each finite type $\mathcal{O}_X$-module on X is isomorphic to exactly one of the $\mathcal{F}_i$.*

Proof. For each open covering $\mathcal{U} : X = \bigcup U_j$ consider the sheaves of $\mathcal{O}_X$-modules $\mathcal{F}$ such that each restriction $\mathcal{F}|_{U_j}$ is a quotient of $\mathcal{O}_{U_j}^{\oplus r_j}$ for some $r_j \geq 0$. These are

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parametrized by subsheaves $\mathcal{K}_j \subset \mathcal{O}_{U_j}^{\oplus r_j}$ and glueing data

$$\varphi_{jj'} : \mathcal{O}_{U_j \cap U_{j'}}^{\oplus r_j} / (\mathcal{K}_j|_{U_j \cap U_{j'}}) \longrightarrow \mathcal{O}_{U_j \cap U_{j'}}^{\oplus r_{j'}} / (\mathcal{K}_{j'}|_{U_j \cap U_{j'}})$$

see Sheaves, Section sheaves-section-glueing-sheaves. Note that the collection of all glueing data forms a set. The collection of all coverings $\mathcal{U} : X = \bigcup_{j \in J} U_i$ where $J \to \mathcal{P}(X)$, $j \mapsto U_j$ is injective forms a set as well. Hence the collection of all sheaves of $\mathcal{O}_X$-modules gotten from glueing quotients as above forms a set $\mathcal{I}$. By definition every finite type $\mathcal{O}_X$-module is isomorphic to an element of $\mathcal{I}$. Choosing an element out of each isomorphism class inside $\mathcal{I}$ gives the desired set of sheaves (uses axiom of choice). $\square$